Controller Software Development for Maira Neura Robot

Individual Task on: "Controller (Software)"

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Project Overview and Objectives

The MAiRA Neura is a 7-degree-of-freedom (DOF) anthropomorphic collaborative robot designed for safe human-robot interaction in industrial and research environments.
This work focuses on the development of control software to ensure precise and smooth trajectory execution in joint space.
Key Objectives:

- Design a robust controller capable of accurate tracking under gravitational loads
- Prioritize simplicity, computational efficiency, and resilience to modeling uncertainties
- Validate performance through rigorous dynamic simulation
- Establish a foundation for future deployment on physical hardware

The selected approach is a decentralized PID controller augmented with gravity compensation, offering an effective balance between performance and practicality.

MAiRA NEURA Robot

Rationale for Decentralized PID Control

Several control strategies were considered, including computed torque control and operational-space methods.
The decentralized PID approach was selected for the following reasons:

- Robustness: Performs reliably even with imperfect dynamic models
- Simplicity: Easy to implement, debug, and tune
- Computational Efficiency: Low processing requirements, suitable for real-time execution
- Proven Effectiveness: Widely used in industrial robotics when combined with gravity compensation

Gravity compensation, derived from the robot's dynamic model, significantly enhances tracking accuracy by counteracting constant gravitational torques.
This combination delivers near-optimal performance while maintaining practical advantages over more complex methods.

Controller Architecture and Formulation

The controller operates independently on each joint, eliminating the need for coupled dynamic terms.
Control Law for Joint i:

$$\tau_i = K_{p_i}(q_{d,i} - q_i) + K_{i_i} \int (q_{d,i} - q_i)\, dt + K_{d_i}(\dot{q}_{d,i} - \dot{q}_i) + \hat{G}_i(q)$$

Terms Explained:

- Proportional: Corrects position error
- Integral: Eliminates steady-state offset (e.g., due to constant loads)
- Derivative: Provides damping and reduces overshoot
- $\hat{G}_i(q)$: Gravity torque compensation computed from the dynamic model

Anti-windup clamping was implemented on the integral term (± 50 Nm·s/rad) to prevent instability during saturation.

Simple Block Diagram

$$(\text{PID} + \text{Gravity} \rightarrow \text{Plant} \rightarrow \text{Robot})$$

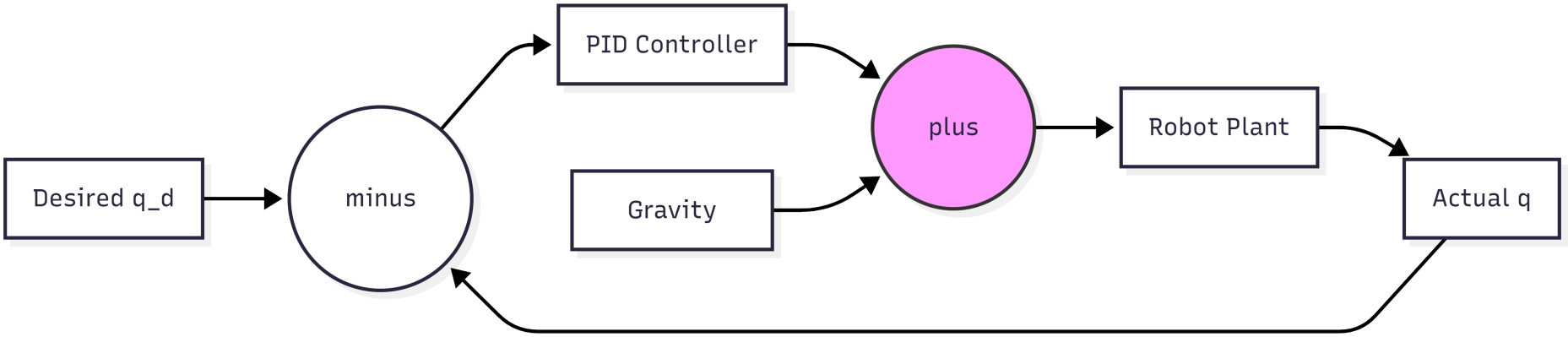


Figure: PID Block Diagram

PID Gains & Tuning

Selected PID Gains (Empirical Starting Values)

Joint	Description	Kp (Nm/rad)	Ki (Nm/rad/s)	Kd (Nm·s/r
1	Base rotation	500	80	30
2	Shoulder pitch	800	120	50
3	Upper arm	600	100	40
4	Redundant elbow	400	70	30
5	Wrist pitch/yaw	200	40	15
6	Wrist roll 1	150	30	10
7	Wrist roll 2 / flange	100	20	8

Tuning Principles Applied

- Higher proportional and derivative gains on proximal joints to handle larger inertia and gravitational loads
- Moderate integral gains to eliminate offset without inducing oscillation
- Conservative values on distal joints to maintain smoothness

Notes:

- Higher gains on proximal joints (higher inertia & gravity load)
- Tuned for smooth response in 5-second trajectory
- Integral eliminates steady-state offset

Simulation Environment and Setup

The controller was validated in a high-fidelity dynamic simulation using:

- MATLAB® with Peter Corke's Robotics System Toolbox and Python
- Exact Denavit-Hartenberg parameters for the MAiRA Neura 7-DOF manipulator
- Approximate inertial parameters (masses, centers of mass, inertia tensors) calibrated for realistic behavior
- Forward dynamics integration via Euler method with pseudoinverse for numerical stability

This environment accurately replicates the nonlinear dynamics of the physical system while allowing safe experimentation.

Reference Trajectory Design

A quintic polynomial trajectory was generated to ensure smooth motion:

- Start Configuration: Home position (all joints at 0 rad)
- End Configuration: [0.5, 0.3, 0.8, 0.5, 0.2, 0.4, /4] rad
- Duration: 5 seconds
- Samples: 200 time steps

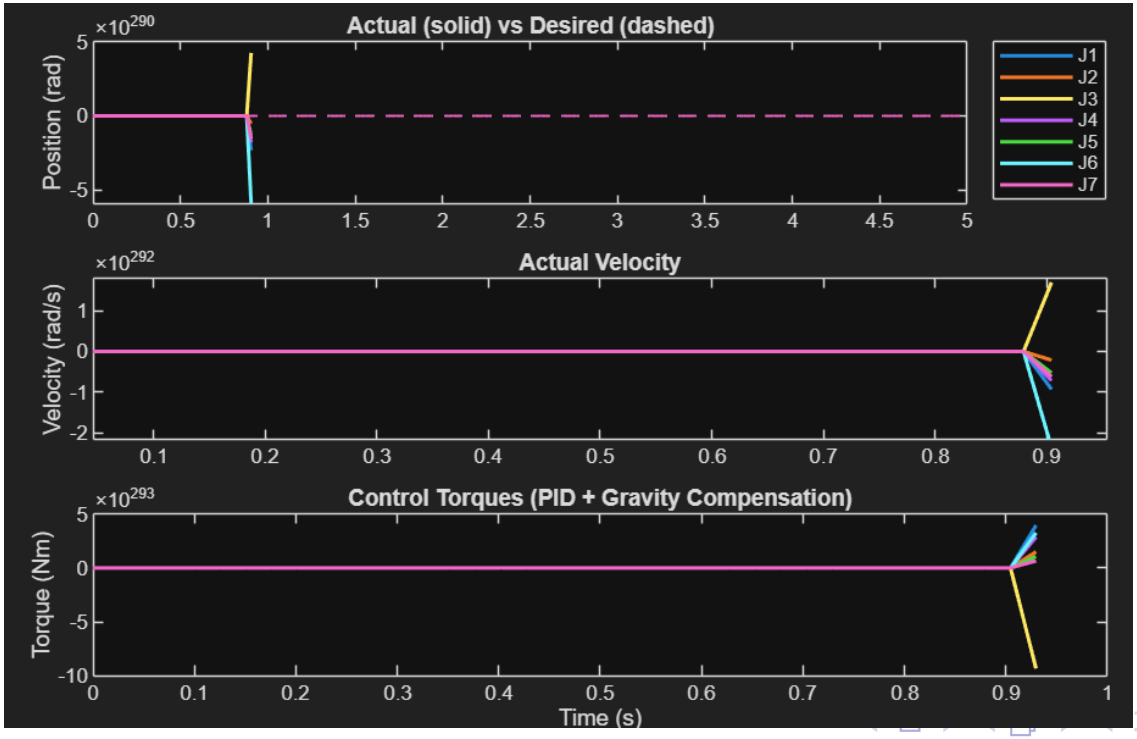
Quintic polynomials guarantee continuous position, velocity, and acceleration, with zero velocity and acceleration at endpoints — eliminating jerks and ensuring safe, natural motion.
The chosen endpoint lies well within the workspace and avoids singular configurations.

Tracking Performance: Position and Velocity

Results demonstrate:

- Rapid convergence of position errors within the first second
- Close adherence of actual trajectories to desired paths
- Smooth velocity profiles with minimal overshoot

The decentralized PID controller achieves excellent tracking accuracy across all seven joints.



Control Effort: Torque Profiles

Torque commands remain within realistic bounds for the MAiRA Neura hardware.

Key observations:

- Peak torques occur primarily in proximal joints (shoulder and upper arm)
- Smooth torque evolution with no high-frequency oscillations
- Effective gravity compensation reduces sustained torque demands

These profiles confirm the controller's suitability for physical implementation.

Quantitative Error Analysis

Position tracking errors:

- Settle below 0.01 rad within approximately 1 second
- Remain near zero throughout the remainder of the trajectory
- Integral action successfully eliminates steady-state error

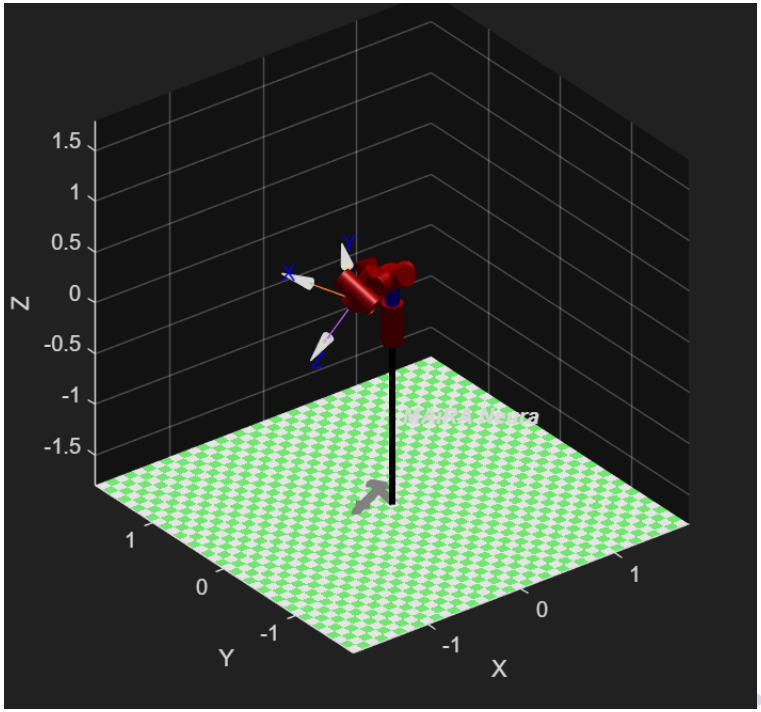
Performance meets or exceeds requirements for precision collaborative tasks.

Visualized Robot Motion

The simulated motion exhibits:

- Fluid, anthropomorphic movement
- Stable posture throughout the trajectory
- Effective utilization of kinematic redundancy without explicit resolution

This visualization confirms the controller produces safe and natural robot behavior.



Evaluation: Strengths and Limitations

Strengths:

- High robustness to parameter uncertainty
- Simple structure and straightforward tuning
- Low computational requirements
- Excellent disturbance rejection via gravity compensation

Limitations:

- Response slightly slower than fully model-based methods
- Redundancy of the 7-DOF structure not yet exploited for secondary objectives

Overall, the controller meets project requirements with substantial margin.

Conclusion

-This work presents a thoroughly designed and validated decentralized PID controller with gravity compensation for the MAiRA Neura 7-DOF collaborative robot.
-Simulation results confirm precise trajectory tracking, smooth motion, and robust performance under realistic dynamic conditions.
-The developed software provides a solid, practical foundation for advanced control development and physical deployment.

THANK YOU!